

Meshlium to automatically send SMS and e-mail alerts based on the sensor data.

Your mobile phone too could be a sensor

Andy Rubin, engineering director, Google, wrote an article on the future of mobiles, based on ideas collated from many experts, and posted it on his blog last year. Therein, he mentions that your mobile phone, if used as a sensor, could reveal tonnes of global patterns and real-time information, very easily.

"Your phone knows a lot about the world around you," he writes. "If you take that intelligence and combine it in the cloud with that of every other

network of environmental sensors that measure pollen, carbon dioxide (CO₂), ultraviolet (UV) sunlight and other atmospheric conditions that can affect human health and daily life. This information will then be provided to companies, healthcare institutions, municipalities, etc, for use in various commercial and educational applications, such as weather forecasting and other information services. The network of sensors appears to be the largest and densest of its kind in Japan.

Gaming console sensor put to different usage

Wii mote is the key control device for Nintendo's Wii console. It is known for its motion sensing and gesture

ing and find out how fast it fell.

In another success story, Willem Luxemburg, a hydrologist at the same university as Hut, demonstrated a water-level sensor made from a Wiimote and a plastic boat. This sensor solves a problem that has intrigued hydrologists for long—measuring evaporation on a body of water. An acceptable, expensive but inaccurate technique followed till date is to place vessels full of water near or atop the water body and put pressure sensors in them. The sensors record the drop in pressure as the water evaporates. You can measure it more easily with a \$40 Wiimote adapted for the purpose.

According to the *Wired Science* report, "The Wiimote could overcome evaporation-measurement problems. It has a tri-axial accelerometer and a high-resolution, high-speed infrared (IR) camera which can sense movement with better than 1-millimetre accuracy. Luxemburg's team tested it in a floating evaporation pan, using a float with an LED. With a Wiimote aimed at the float, and some hacking and programming of the Wiimote's output, they were able to get highly accurate, real-time data on water level wirelessly sent to a laptop. The IR camera can track up to four LED lights at once, so scientists can use several floats to calculate the water's plane. To be as accurate with pressure sensors, you would need more and costlier units."

Sensors that consume less power

Apart from the several advantages, wireless networks have one significant disadvantage too over their wired incumbents—these consume more power, increasing the total cost of ownership of the sensor network in the long run.

A patent-pending media access protocol developed by researchers at the University of California at Berkeley might be the solution to this problem. This protocol is suitable for networks where the sensor nodes are power-constrained but the network's main access

"Globally, wireless sensors are popularly used for military, medical and environmental applications since these are the most critical areas that need to be addressed"

phone, we have an incredible snapshot of what is going on in the world right now. Weather updates can be based on not hundreds of sensors, but hundreds of millions. Traffic reports can be based not on helicopters and road sensors, but on the density, speed and direction of the phones (and people) stuck in the traffic jams."

Looks like somebody has decided to try this out too! The National Science Foundation, USA, has awarded a \$1.5-million grant to CitiSense, a cell-phone-based sensor network system, to develop and deploy hundreds of small environmental sensors that can be carried by the public in San Diego. These sensors will use cellphones to relay data. In addition to environmental sensors, the people can volunteer to wear biological monitors to collect health-related statistics. The information will then be analysed using computing systems and relayed back to the people.

In November 2009, telecom service provider NTT DOCOMO Inc. announced the trial operation of a

recognition capabilities. It uses an accelerometer and optical sensor technology to allow users to interact with the on-screen items through gestures or by pointing their fingers.

This device has fired the imagination of many a hacker. Last year, a developer called Johnny Chung Lee turned it into a finger-tracking device and a touch-screen whiteboard. You can do it too, thanks to his do-it-yourself instructions available at <http://johnnylee.net/projects/wii/>.

Inspired by this, physicist Rolf Hut of the Delft University of Technology has built a Wiimote wind sensor. In a *Wired Science* report, Hut comments, "It was just a bendy pole with an empty bottle on top with an LED light on the bottle." It was built in a single day, and a little soldering is all it took.

If the report is to be believed, there is nothing you cannot sense with a Wiimote and an LED light. Of course, it might not be the best way of sensing a parameter, but you can definitely find a way out with it. For example, you can put a Wiimote atop a collapsing build-